

Project Executable Files

Date	02 July 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Project Executable Files

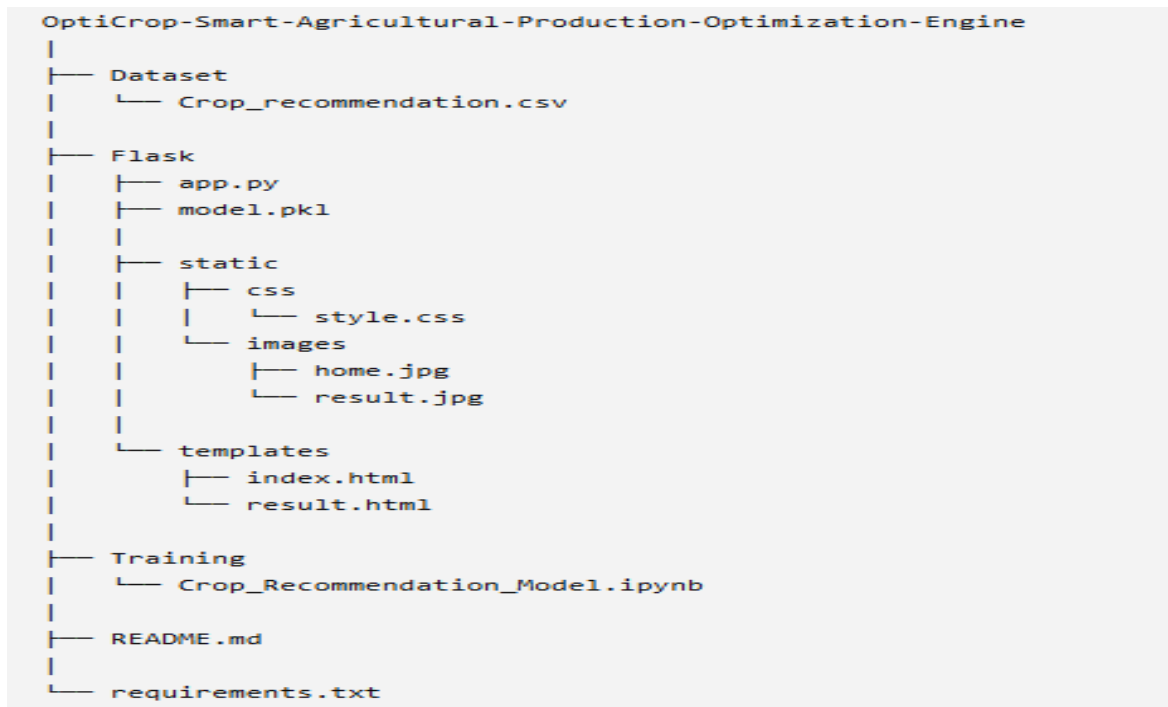
This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example)	No
6	Deployed application URL (if hosted)	No
7	APK / Executable binary (if applicable)	NA
8	Dockerfile / Containerization configuration	NA
9	Test files and test results	Yes
10	Demo video / Walkthrough	Yes (if prepared) / No (if not prepared)

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:



Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not Hosted
Login Credentials (Demo)	Not Applicable
Platform / Hosting Provider	Local System
Repository Link	https://github.com/Charishmakongara/OptiCrop-Smart-Agricultural-Production-Optimization-Engine
Demo Video Link	https://drive.google.com/file/d/1O7trT-WaYMTMkvGK81GIWwS4O7v46pO6/view?usp=drivesdk

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

Step 1

Clone or download the project repository.

Step 2

Open the project in **Visual Studio Code**.

Step 3

Navigate to the **Flask** project folder.

Step 4

Install all the required Python libraries using the **requirements.txt** file.

Step 5

Run the Flask application using the **app.py** file.

Step 6

Open the application in a web browser using the local Flask server.

Step 7

Open the **Crop Recommendation** page.

Step 8

Enter the following details:

- Nitrogen (N)
- Phosphorus (P)
- Potassium (K)
- Temperature
- Humidity
- pH
- Rainfall

Step 9

Click the **Predict** button to generate the crop recommendation.

Step 10

The application displays the **recommended crop** based on the entered soil and environmental parameters.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Application requires manual input of soil parameters.	Users must enter accurate values for better predictions.
2	Model is trained on the available dataset only.	Performance can be improved by using larger and updated datasets.
3	Weather API and fertilizer recommendation are not integrated.	These features can be added in future enhancements.